INCH-POUND

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COMMERCIAL ITEM DESCRIPTION

SUPPLIED AIR RESPIRATOR WITH BACKUP
SELF-CONTAINED BREATHING APPARATUS (SAR/SCBA)

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE.

1.1 Scope. This commercial item description covers the Supplied Air Respirator with a backup Self-Contained Breathing Apparatus (SAR/SCBA) for use in providing breathing air for entry into and escape from spaces aboard Navy ships where atmospheres are classified as being Immediately Dangerous to Life or Health (IDLH). The SAR/SCBA is primarily intended for use in Gas-Free Engineering applications. This commercial item description does not cover the SAR external air source equipment.

2. SALIENT CHARACTERISTICS

2.1 Physical and performance characteristics. The SAR/SCBA shall operate in the pressure demand mode. The SAR/SCBA shall carry the approval and certification of the National Institute for Occupational Safety and Health (NIOSH) for an open circuit, pressure demand, combination SAR and SCBA. The SAR/SCBA shall meet the NIOSH design, construction, and functional requirements in 42 CFR Chapter 84, subparts H & J. In the event of a conflict between this specification and the NIOSH requirements, the more restrictive requirement shall take precedence. The SAR/SCBA must be capable of satisfactory airline operation through a hose or multiple connected hoses with a minimum length of 250 feet between the external air supply source and the SAR. The required external air supply source pressure for the SAR shall not be greater than 125 pounds per square inch gauge (psig). In the event that the supplied air from the external air supply source is interrupted, the user shall be provided with the capability to readily switch to the backup SCBA located in an over-the-shoulder mounted unit.

2.1.1 Facepiece. Each facepiece shall be a full mask and have an adjustable head harness. The facepiece shall permit the optional use of NIOSH approved corrective spectacles without compromising the protective features of the apparatus. The facepiece shall also be designed to minimize fogging of the eyepiece and spectacles. An adapter shall be provided with each facepiece to permit temporary attachment of an air breathing cartridge for use during fit testing of the facepiece in the negative pressure mode in accordance with Positive Pressure Respirators of ANSI Z88.2. The facepiece shall be fully immersible in a dilute solution (2 tablespoons per 1 ½ gallons of water) of Wescodyne G or sodium hypochlorite, for the purpose of routine disinfecting and cleaning.

2.1.2 Second stage, low pressure regulator. A second stage, low pressure regulator shall be a pressure demand type regulator and shall maintain a static positive pressure in the facepiece of not greater than 1.5 inches water gauge (w.g.). The pressure in the facepiece shall not fall below atmospheric during inhalation at air flow rates of less than 4 cubic feet per minute (cfm). An exhalation rate of 3 cfm shall not exceed the static pressure in the facepiece by more than 2 inches w.g. A don/doff switch/push button shall be provided on the second stage, low pressure regulator as a convenient means to start and stop the flow of breathing air to the facepiece when donning and doffing the facepiece. If any part of the second stage regulator is exposed to the exhalation path, the second stage regulator shall be fully immersible in dilute solution (2 tablespoons per 1 ½ gallons of water) of Wescodyne G or sodium hypochlorite for the purpose of routine cleaning and disinfecting.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navsea.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.1.3 Breathing air hose(s). The SAR/SCBA shall be equipped with two 75-foot lengths of breathing air line hose to connect the SAR/SCBA to the external air supply. The working pressure of the air line hoses connecting to the external air supply source shall be rated for at least 125 psig. All external air supply hoses shall be fitted with quick-connect fittings consisting of Foster type SL 3003 W "Safety Lock" quick-connect female couplings and Foster type 11-3S/S stainless steel male coupling plugs to interface with the Navy's supplied air systems. Separate breathing air hoses shall be supplied to connect the facepiece to the backup SCBA air supply. Hoses shall be tested to twice working pressure and have a permanently attached tag with the test pressure and test date.

2.1.4 Backup SCBA. The backup SCBA shall be rated by NIOSH as providing a 15-minute service life of breathing air. The backup SCBA shall meet the service life requirement after stored for 6 months in the environmental conditions specified in 2.1.8 without being replenished.

2.1.4.1 Backup SCBA assembly. The backup SCBA assembly shall be made from a durable fabric/material and shall have adequate construction to protect the composite air cylinder(s) from being scratched or otherwise damaged when the fully loaded unit is slid across the flat edge of a steel angle or channel. The wearable weight of the fully loaded unit (including straps/belts) shall not exceed 13.0 pounds. The backup SCBA unit shall be designed with a shoulder strap to allow it to be carried over the shoulder and shall contain the backup SCBA air cylinder(s), the backup SCBA cylinder pressure regulator(s), the backup SCBA pressure indication device, the backup SCBA low pressure alarm, and the backup SCBA air supply control valve(s). The backup SCBA unit shall also include a waist belt to fasten the unit to the body of the user. The shoulder strap and waist belt shall be constructed of a nylon fabric and shall be at least 2 inches wide.

2.1.4.2 Backup SCBA air cylinder(s). The backup SCBA air cylinder(s) shall be fully wound fiber-wrapped composite construction and certified under current requirements established by the Department of Transportation (DOT), and shall be rated for a breathing air working pressure not greater than 4500 psig. Cylinders shall be delivered fully charged with Type 1, Grade D quality air in accordance with CGA G-7.1 and with a maximum moisture content of 20 mg/m^3 or a dew point less than or equal to -65 °F (-54 °C).

2.1.4.3 Backup SCBA pressure indication device. The backup SCBA shall include a pressure indication device that can be read by the user with the facepiece donned under the range of lighting conditions specified in 2.1.8.

2.1.4.4 Backup SCBA low pressure alarm. The backup SCBA shall include a low pressure audible alarm, which sounds continuously when the backup cylinder pressure has decreased to a pressure which is 20% to 25% of the cylinder's rated pressure. The alarm shall be audible under the range of conditions specified in 2.1.8.

2.1.4.5 Backup SCBA air supply control valve. The backup SCBA shall include an air supply control valve(s) to permit manual opening/closing of the backup SCBA air cylinder(s). The cylinder valve shall utilize a CGA 346 or 347 connection.

2.1.5 SAR/SCBA hard shell carry case. The SAR/SCBA shall be provided with a hard shell carry case. The case shall have plastic and/or corrosion resistant metal hardware that is resistant to a saltwater environment. The case shall be plastic in construction, suitcase in style, orange, and have a centered handle for ease of carrying. The length of the case shall be measured along the opening axis and shall not exceed 25 inches. The depth of the case shall be measured on an edge perpendicular to the edges formed by opening the case and shall not exceed 9 inches. The total internal volume of the case shall not exceed 3400 cubic inches. No visible damage or functional degradation of the case or its contents shall result when the fully loaded case is dropped on its shortest edge from a height of 2.0 feet onto a steel deck. The case shall contain the SAR/SCBA facepiece with second stage regulator, one 75-foot section of breathing air line hose, the backup SCBA assembly with its included equipment, a copy of the operation and maintenance manual, and the spare parts kit.

2.1.6 SAR/SCBA weight. When the air supply hose from the external air supply source is disconnected from the SAR/SCBA, the fully charged SAR/SCBA weight borne by the wearer shall not exceed 15.0 pounds.

2.1.7 Storage service life. The SAR/SCBA shall have a minimum service life of 14.5 years.

2.1.8 Environmental conditions. The equipment will be stored and operated onboard U.S. Navy vessels. The equipment shall be operational in temperatures ranging from –25 °F (-32 °C) to 205 °F (96 °C) and when stored at temperatures ranging between -4 °F (-20 °C) and 149 °F (65 °C).

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2.2 Design and construction.

2.2.1 Materials. Materials shall be durable and corrosion resistant. Any lubricants used shall be nontoxic and suitable for high pressure breathing air systems.

2.2.2 Interchangeability. Part and component interchangeability shall be inherent in the fabricated products. All parts and assemblies assigned a unique part number shall be functionally and physically interchangeable.

2.2.3 Maintainability. The equipment components shall be modular in nature and feature standard parts to facilitate repair.

2.2.4 Safety. Assembled units and components shall eliminate hazards to personnel, interfacing equipment, and the equipment itself. The equipment shall be safe to use in the intended environment as specified herein and shall include over-pressurization protection from the supplied air source and over-pressurization protection of the backup SCBA cylinder.

2.2.5 Workmanship. Mechanical components of the equipment shall be uniform in appearance and free from cross-threading, burrs, scratches, cracks, breaks, pits, dents, chips, sharp projections or edges, loose parts, or foreign material that may adversely affect equipment serviceability, performance, reliability, safety, endurance, or wear.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Inspection requirements. Bid samples may be required as specified (see 6.3).

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES

6.1 Source of documents.

6.1.1 ANSI. ANSI standards may be obtained from the American National Standards Institute, 25 W. 43rd St, 4th Floor, New York, NY 10036 or online at http://webstore.ansi.org/.

6.1.2 CFR. The Code of Federal Regulations may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20401 or online at www.gpoaccess.gov/index.html.

6.1.3 CGA. CGA specifications and standards may be obtained from the Compressed Gas Association, Inc., 4221 Walney Road, 5th Floor, Chantilly, VA 20151-2923 or online at www.cganet.com.

6.1.4 FAR. The FAR may be obtained from the Regulatory Secretariat, 1800 F Street NW, Washington DC 20405 or online at www.acqnet.gov.

6.2 National stock numbers (NSNs). The following is the NSN that is assigned to this CID. It may not be indicative of all possible NSNs associated with the CID.

4240-01-440-6605 SAR/SCBA System

6.3 Ordering data. The contract or order should specify the following:

a. CID document number and revision.

b. Product conformance provisions.

c. Packaging requirements.

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6.4 Key words.

IDLH

Immediately Dangerous to Life or Health

PASP

Primary Air Supply Pack

RASP

Reserve Air Supply Pack

SAR/SCBA

MILITARY INTERESTS

Custodians:
Army - EA
Navy - SH
Air Force - 03

Preparing Activity:
Navy - SH
(Project 4240-0700-000)

Review Activities:
Navy - AS
Air Force - 11, 84, 99

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